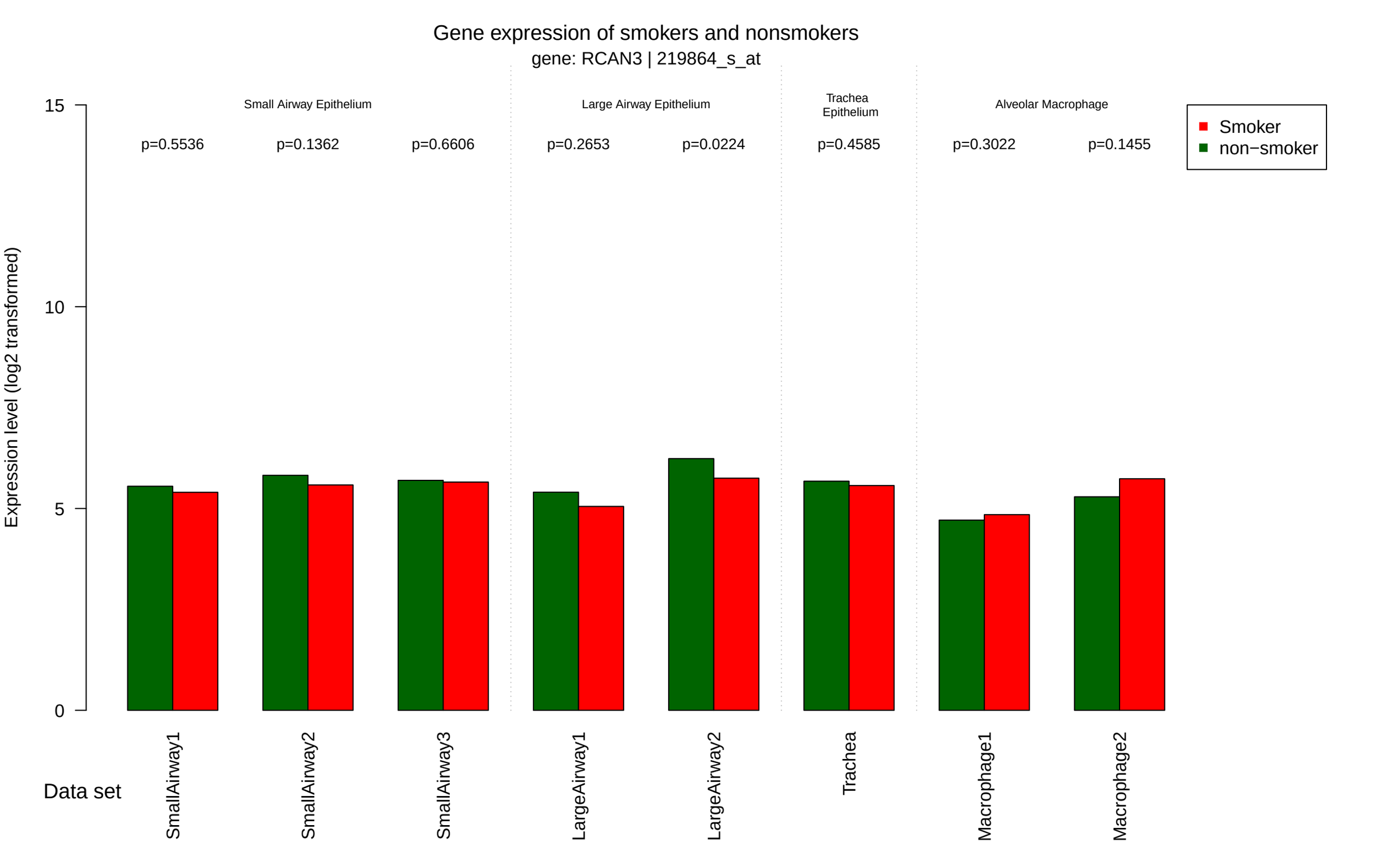
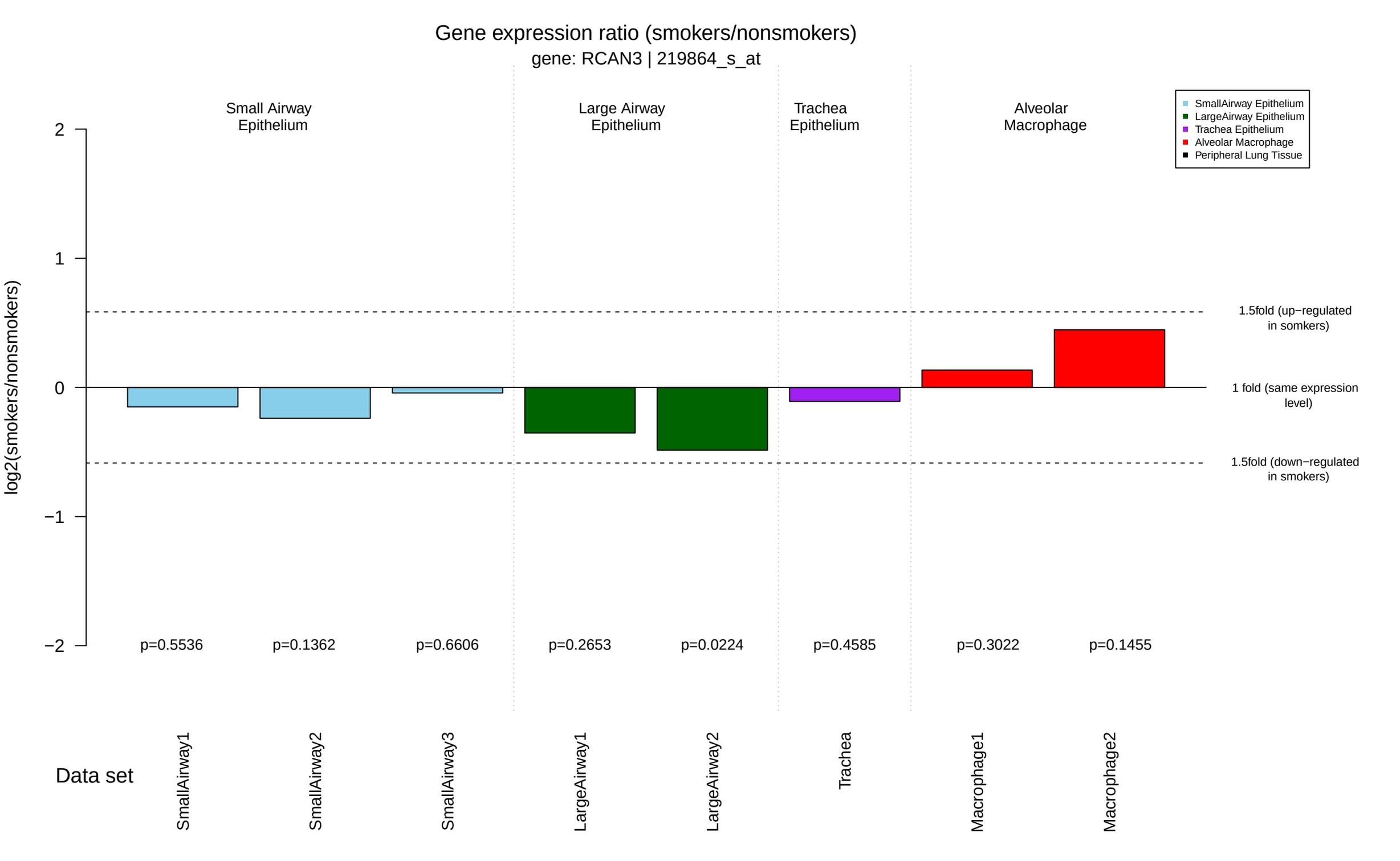
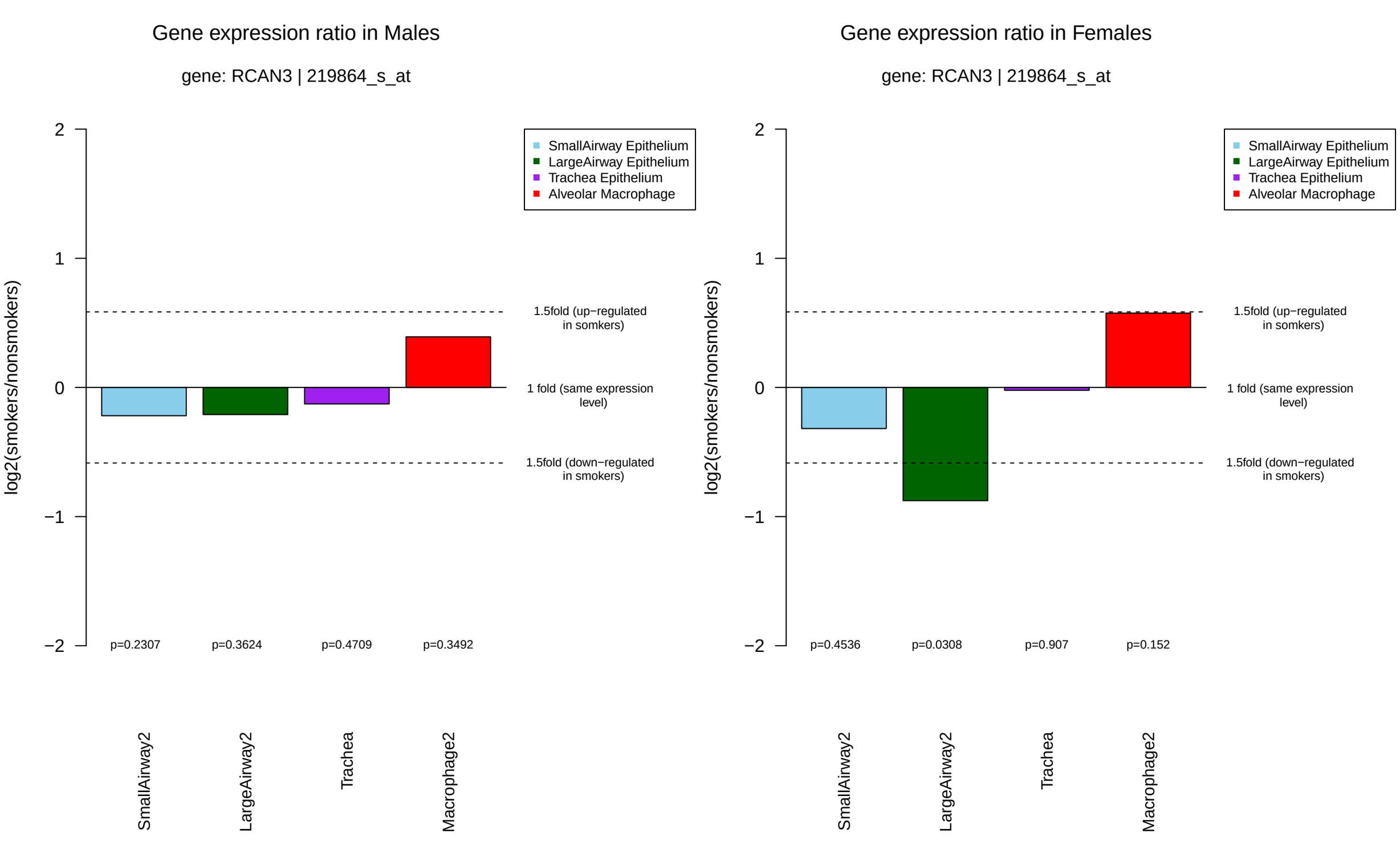
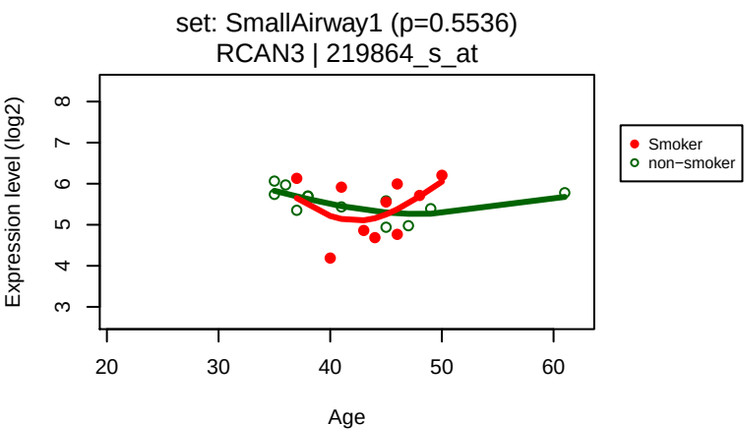
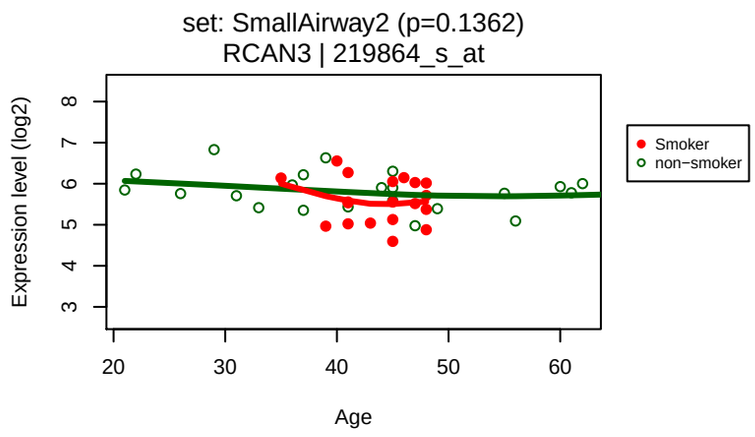
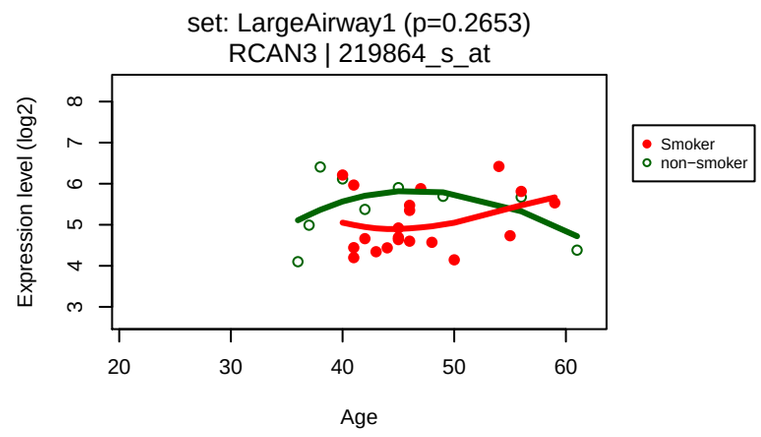

Gene expression of smokers and nonsmokers
gene: RCAN3 | 219864_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5536
p=0.1362
p=0.6606
p=0.2653
p=0.0224
p=0.4585
p=0.3022
p=0.1455
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

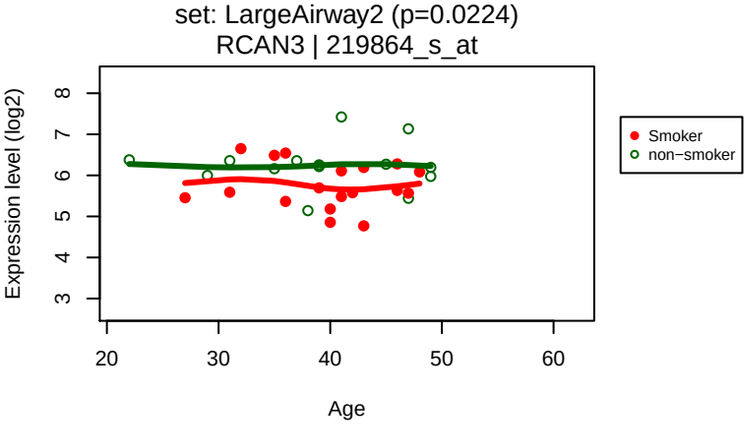

Gene expression ratio (smokers/nonsmokers)
gene: RCAN3 | 219864_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5536
p=0.1362
p=0.6606
p=0.2653
p=0.0224
p=0.4585
p=0.3022
p=0.1455
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

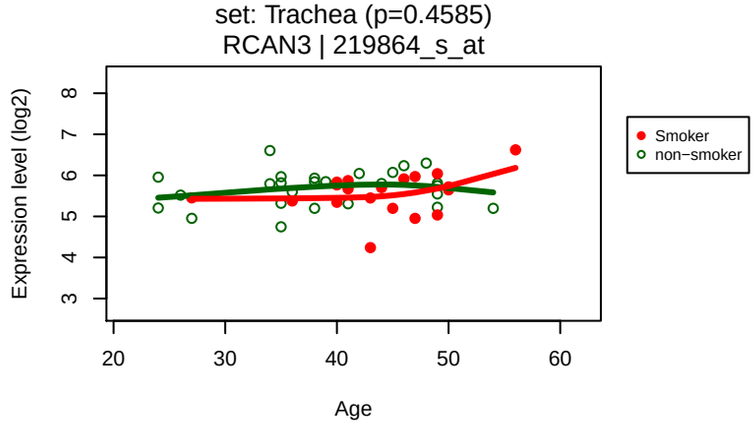

Gene expression ratio in Males
gene: RCAN3 | 219864_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2307
p=0.3624
p=0.4709
p=0.3492
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCAN3 | 219864_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4536
p=0.0308
p=0.907
p=0.152
SmallAirway2
LargeAirway2
Trachea
Macrophage2

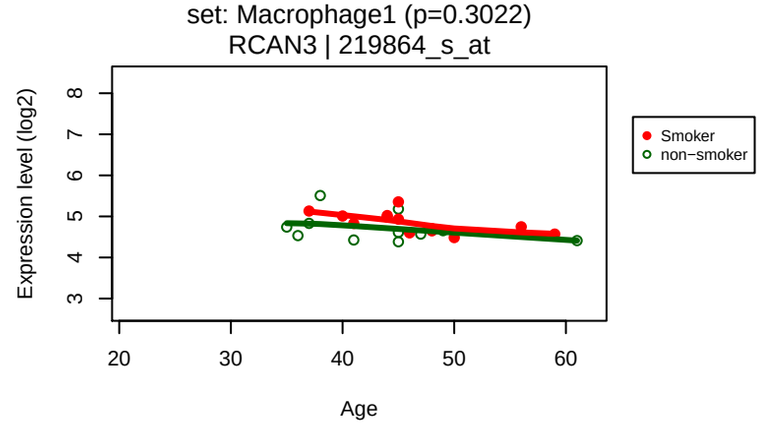

set: SmallAirway1 (p=0.5536)
RCAN3 | 219864_s_at
Expression level (log2)
Age
Smoker
non−smoker

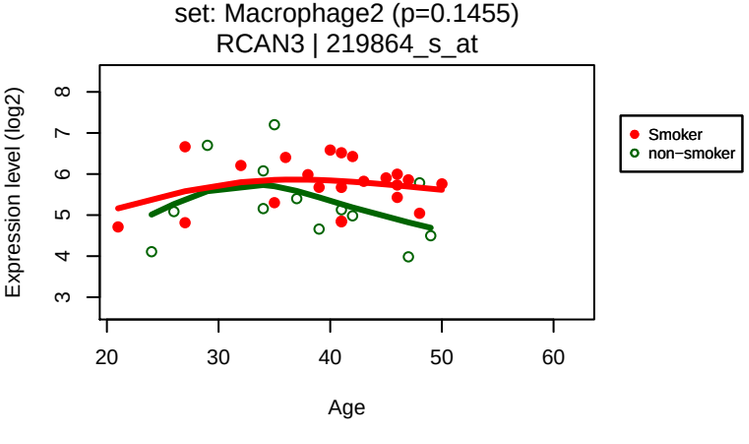

set: SmallAirway2 (p=0.1362)
RCAN3 | 219864_s_at
Expression level (log2)
Age
Smoker
non−smoker

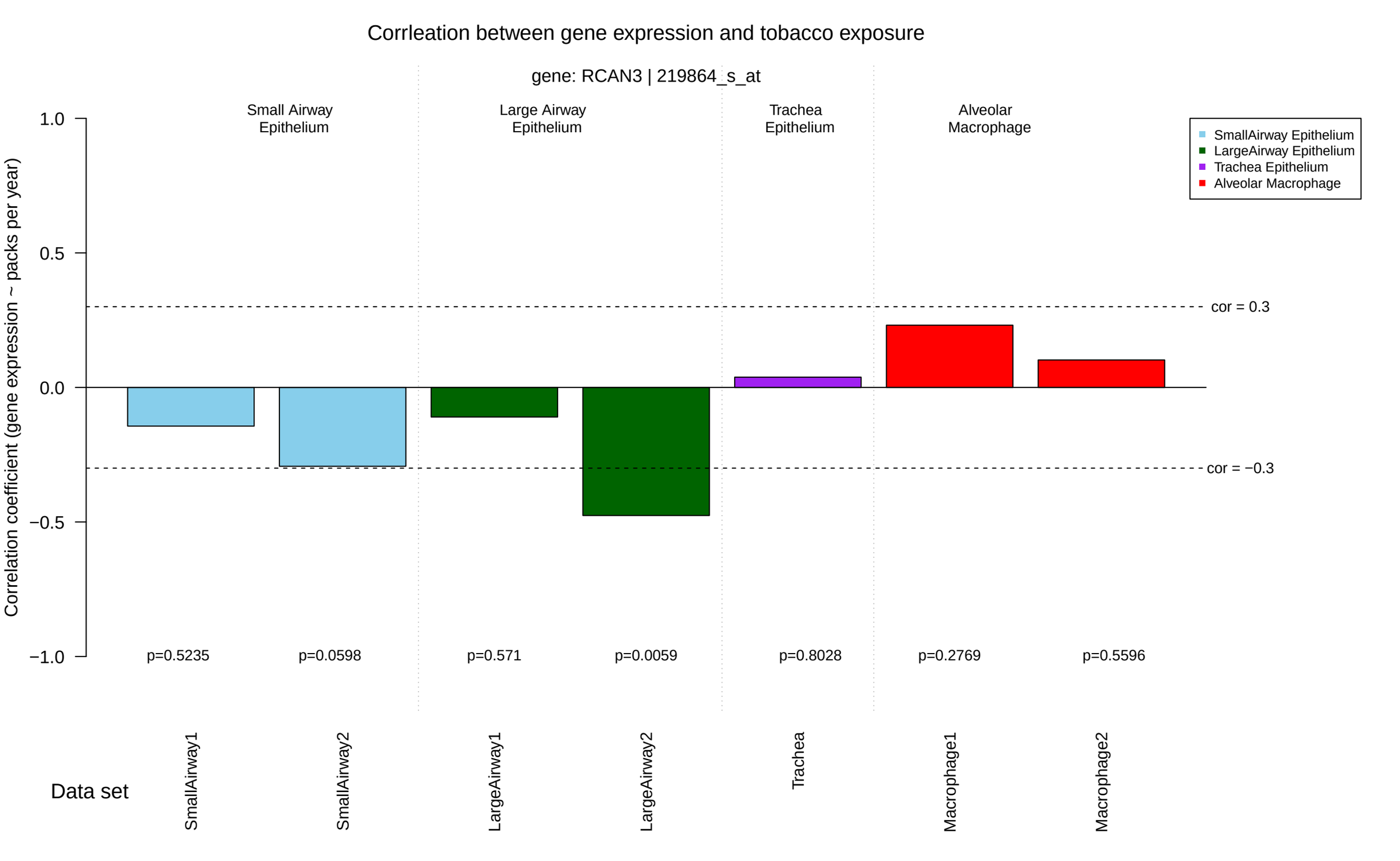

set: LargeAirway1 (p=0.2653)
RCAN3 | 219864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0224)
RCAN3 | 219864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4585)
RCAN3 | 219864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3022)
RCAN3 | 219864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1455)
RCAN3 | 219864_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCAN3 | 219864_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5235
p=0.0598
p=0.571
p=0.0059
p=0.8028
p=0.2769
p=0.5596
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2